

***Self-Designed Role-Playing Game and Multi-Agent System
to Empower a Local Land Use Management Decision-Making Process:
The SelfCormas Experiment in Senegal***

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Self-Designed Role-Playing Game and Multi-Agent System to Empower a Local Land Use Management Decision-Making Process: The SelfCormas Experiment in Senegal

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I. A novel form of a Role-Playing Game to accompany a collective decision-making process in land use management.

Role-Playing Games (RPGs) have already been used to support land use management (Commere 1989, Mermet 1991 et 1993, Piveteau 1994, Barreteau *et al.* 2001). The games, which depend upon the prior diagnosis of the situation by experts, help players to share in this analysis and to draw upon some improvements from it. But in our own experiment, we use a decidedly different form of RPG: a Self-Designed RPG. There are no prior rules and the setting of the game is drawn from our own previous diagnosis. The RPG is solely designed from a self-analysis of the players' situation and is produced in a single step—without the players' knowledge that this analysis will afterwards be used to set up a RPG.

Necessarily, this type of RPG relies upon very simplified rules and a crude game setting. But the goal is not to conceive of a game that is technologically advanced and satisfying from our own expert point of view. Our purpose is to create a novel design that includes "mediating supports" (Boltanski and Thévenot 1991), which could facilitate the emergence of worthwhile debates by taking into account all the different perspectives and building upon some truly communal technical devices (D'Aquino *et al.* 2001 et 2002). The method seeks to help people to progressively formalize the elements—as they move forward through their debates—that appear useful to us all for the improvement of their decision-making abilities. In fact, the usual technical complexity that designers often put into a RPG designed to support negotiation is due to the designers' need to reproduce any possible impact, especially an *environmental* impact, of the interactions between any players. This type of designer's RPG must take into account all the possible dynamics and effects that are potential between the environmental and socio-economical elements in order to help decision-makers to find the best solution. This sort of designer instinctively contrives the whole process in the form of a perfect decision being found in a given timeframe (the game's time), as opposed to the realistically complex situation that he should endeavor to reflect in his RPG. His tacit hypothesis is that all the essential elements for the decision can be selected prior to the game and then deposited in the RPG. Moreover, he places himself in the role of an expert who proposes to the players his own, more perfect external perception of the situation as an ideal support for the improvement of the decision-making process. In fact, even if there is minimal participation by the people effected, this sort of formalized modeling is always a personal work of the designer; all the more so since a major concern for the designer is often the collection of a maximum of diverse data for carrying out an analysis which will be the closest possible to reality. Yet this entire process infers that the goal is to *produce* a tool

that will be able to guarantee to the decision-makers its conformity with their reality. The risk lies in the difficulty of reaching a model sufficiently close to the complex nature of the reality without falling into such an intricate device that it is no longer suitable to use efficiently. The possible weaknesses of this model lie in the ambiguous character of the designing expert, who often positions himself as the master of the decision process. Many information systems fall into these traps.

At the outset, we believe the decision-making process is primarily a continuous, iterative and incremental process. This means that it will always produces imperfect decisions, yet progresses towards less imperfect ones. So the purpose of an effective tool of this type cannot be to *produce right decisions* but to help people to improve *their own imperfect decisions*. Hence, the degree of complexity of the decision-making tools should mirror the progressive, continuous and iterative nature of the decision-making process, and this is the first challenge in the creation of supporting tools. Next, we think the complicated reality—particularly the complexity of the interactions, which are too multifarious to be simplified—should be taken into account in land use and natural resource management issues, which are never as neutral as designers would like to believe. Any austere synthesis of such a complex matter is a point of view that confuses the material elements involved with personal ideas about the "right ways" of managing resources, land, and eventually society. That is why we then believe that an accurate and non-neutral selection among alternatives should be taught directly to the decision-makers, i.e., the stakeholders, local representatives and politicians. This approach would be more efficient since it will more closely reveal the points of view and therefore the needs of decision-makers, and it would also be more legitimate since local democracy should not be an arena where experts and people "share"¹ the decision-making authority.

Consequently, we tackle here the design of decision-making tools in a different manner. In effect, contrary to G.D. Brewer (1985), we do not believe that the fostering of the internal representation of the complexity involved has a lower operational value than classical experts' analyses. First, this internal construction will reduce the non-use of tools, often due to the placement of the design over the needs they are supposed to meet. Second, the tool will be most efficient when its design speaks closely to the "real matter" (i.e., the matter such as it appears to decision-makers), than when its design addresses the "environmental matter" (i.e., the matter such as it appears to exogenous people). What is the progress in decision-making in this case? The improvement in the process depends upon providing communal supports that help decision-makers in their internal debate and learning-by-doing approach. We believe that the main constraint in land use management issues is not dependent upon technical advice or knowledge but in the efficiency of the socio-political dynamic of debate, decision and action. A collective, effective, and sustainable socio-political process, in which tools merely serve the interests of stakeholders and their representatives, is necessary for tackling land use management. The first aim of our supportive tools is to accompany the decision-makers in their own iterative and progressive route towards a communal perception, then decision, concerning the complex matters they face. So we do not necessarily need sophisticated tools to

¹ Cf. the ambiguity of "participatory" approaches (D'Aquino 2001).

develop an efficient decision-making process in a complex environment; in any case, a complex formalization is always the wrong kind of simplification.

All this leads to our *self-design* principle (D'Aquino *et al.* 2001 et 2002): the more endogenous the design is, the more appropriate the tool is. The challenge is therefore to conceive of a methodological framework that places at the disposal of the stakeholders and others principals all relevant knowledge and information so that they themselves can manage their issues. Expert knowledge and appropriate information may thus be summoned up progressively by the principals upon their own request and within the context of their own objectives. That is, tools able to (i) take into account their own perceptions; (ii) put advisers' knowledge at their disposal; (iii) be directly controllable by them. This is how we experimented in Northern Senegal.

In Senegal, our team, along with the local principals, is focused on tackling multipurpose land use management issues and solving the puzzle of sustainable development. Our aim is to help these principals to progress towards an improved self-management, especially by vigorously involving the stakeholders. In other words, this experiment has engaged in a practical way our above-mentioned hypotheses: taking into account people's own perceptions, putting external knowledge at their disposal, and encouraging the direct supervision of every step by them. Accordingly, we have tested a methodology for the shared collective design of planning supports that will help the local principals to enhance their empowerment in planning decisions regarding sustainable land-use management. This "self-design" experiment was organized in the form of discussion workshops directed toward a primary land use scale of around 2,500 km² and 40,000 people. Several different experiments were organized throughout the span of a year (see map in Illustration 1), beginning with a self-designed RPG as well as a self-designed computer modeling.

II. Accompanying the self-design of a RPG for communal land use management.

The first step of our method is the identification and then acknowledgement² by the participants (stakeholders, representatives and principals) of the *satisfaction criteria* for the various kinds of stakeholders. The *satisfaction criteria* have been explained as "the fundamental elements for every stakeholder to succeed in providing a living for his family". This sort of "gentlemen's agreement" is essential; it allows shifting the analysis towards elements no one would contest and disposing of conflicting matters by harking back to the deeper incentives behind them.

² In other words, this comprises the definition, the collective approval and ultimately the *legitimization* (cf. patrimonial approach: de Mongolfier and Natali 1987, Ollagnon 1989 et 1993, Mermet 1993) of the minimum requirements of each stakeholder.

For instance, in the first experiment³, participants had agreed that:

- Farmers needed a spring nearby, not more than 500 m away, and a piece of land not liable to flooding.
- Breeders needed an access to a fresh watering place nearby, not more than 800 m from grasslands; and the grassland, which should be no more than 800 m from stations, should flood every year but not for more than 20 days.
- Fishers needed a flooding zone not more than 800 m from every settlement and within certain conditions (with and without vegetation, different times of flooding, etc.).
- Hunters (wild geese and ducks) needed a flooded zone until March (i.e., for six months).
- The needs of parks are represented by the needs of its main residents: wetlands migratory birds. These fowl have been deemed to need various watering zones (with and without vegetation, shallow and deep places, etc.) free from disturbance by the closest human residents.

The second step of our method consists of identifying *during the course of time*, throughout the entire year, the same *satisfaction criteria* for each activity. One can see a demonstration of this step in the chart below in Illustration 2. This detailed synopsis is the kind of highly detailed support that allows the formation of an accurate communal representation of the matter and therefore an operationally successful decision process. Nonetheless, it is a self-designed chart, designed and filled out solely by participants, i.e., the stakeholders and their principals (see Illustration 2).

This communal acknowledgment is a real collective initialization of the particular situation of every type of stakeholder, their incentives, their constraints and their difficulties. This shared knowledge and agreement are together the first milestones we reach and provide a solid footing for the future success of the decision-making process. No external input, no technical knowledge, no expert advice: merely information perhaps already known by every participant but not yet communally constructed in a public arena and presented without arousing resentment at anyone's behavior or actions. This is the goal of the first phase of our method: an endogenous construction of a situation sufficiently close to actual circumstances to design a RPG that will permit us to examine the space and time compatibilities among the different land uses.

As an illustration, one can see some charts from the fourth workshop ("Gnith" zone). Here, participants defined only two types of stakeholders (breeder and farmer) that were

³ Outlying zone of Djoudj Birds National Park, Senegal.

relevant to the main question they had agreed to focus on: competition between agriculture and breeding. For farmers, the classification was the following:

"Best place" for agriculture.

Soil	Rice culture	Gardening
False <i>hollaldé</i> (vertic soil)	9	2
<i>Hollaldé</i> (clayey)	10	1
<i>Fondé</i> (silt and clayey)	8	6
<i>Deck</i> (sandy and clayey)	7	3
<i>Deck dior</i> (silt and sandy)	7	10
<i>Dior</i> (sandy)	0	9

"Best place" for breeding.

Location	RainySeason	Beginning of Dry Season	After crops
<i>Diéri</i> (sandy)	10	10 (if less than 5 km from a watering place)	10 (if less than 5 km from a watering place)
Irrigated crops	inaccessible (under cultivation)	inaccessible (under cultivation)	9
Temporarily flooded zones	inaccessible (water)	inaccessible (water)	Between 2 and 10 (according to the duration of flooding)
Drainage zones	inaccessible (water)	inaccessible (water)	2
Shore of Guiers lake	inaccessible (under cultivation)	inaccessible (under cultivation)	6

The goal of the second phase is to then allow participants to "set in motion", and therefore to test, their first communal representation of the land use management issue that has previously been taken up. The self-designed RPG becomes this "setting in motion", in which every stakeholder described before is one of the characters. Indeed, as seen in the previous Chart, one notices people selecting during the participatory analysis, not only spatial and temporal information about land uses, but also the types of stakeholders they believed it was important to take into account (fisher, farmer, breeder, etc.). For every stakeholder, participants identified the resource needs of each type, including for example the matter of distance. We were then able to organize a RPG from

these elements to help participants to simulate the scenario that they had previously designed: a *role playing game self--design*, i.e., a RPG designed solely through the process of their own analysis. The different types of stakeholders selected by participants were the players of the game and the pertinent elements were merely drawn from their previous debates and served to frame the game; its basic rules were drawn from the Charts seen in Illustrations 2 and 3; and its setting derived from cartographic support based on previous charts. Each player must attempt to satisfy the needs required by his character throughout the yearly cycle. Hence, month after month in this game, each player decided on his activity and his position, according to a definition of his requirements, which were previously established jointly (see Chart below). The scores given in the charts above for each type of resource led to a rough quantification of these requirements.

For instance, in the case of the first workshop ("Djoudj site"), the game was as follows:

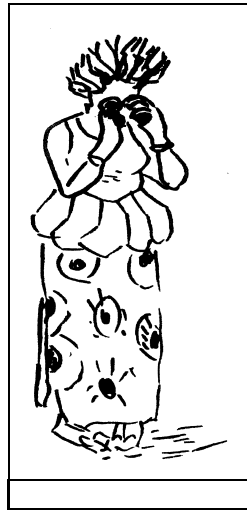
1. The cartographic support (see Map in Illustration 4) enlisted use of the previous endogenous legend from the first steps (see Chart in Illustration 1) and used a GIS to fill out the map; two irrigation canals have been added, to reconstruct a truly typical situation.
2. To show his position at each time interval (one month), each player uses a post-it note as a token that he moves around the map (see the first two Pictures in Illustration 6).

Stakeholder figurines chosen and then used by participants in "Djoudj" Workshop.			
Farmer	Breeder	Fisher	Migratory Bird
			

3. The cartographic support was divided roughly equally into compartments for more convenient linkage between users and resources (see Map in Illustration 4).
4. Six types of user tokens were distributed among 13 players: 4 farmers, 1 small breeder, 2 big breeders, 2 fishers, 1 hunter, and 3 migratory birds.

5. At the beginning of the game, each participant drew lots for a character (in fact, a type of resource user), but no participant could play the character he was in reality.
6. "Sanctions" were given to players who had not succeeded in satisfying the needs of their character. A drawing of a crying woman, which symbolizes a family in need, depicted this sanction.

**"Sanction" figurine:
The needs of the user have gone unsatisfied during a yearly cycle.**



7. A single turn in the game lasted one month, and the game itself began after the rainy season ended (October). Every player moved his character as he liked upon the map at each turn. If adequate resources existed at the spot, then afterwards, when the scoring of the game occurred (see just below), the user would be considered satisfied.
8. At the end of the yearly cycle, participants publicly took stock of the sanctions and "fines" (when a player did not respect a rule) for each player. Then, one after the other, each player explained the motives for his behavior and the difficulties during the recently past cycle. This is an enormously important step, which allows a communal and shared analysis of the various interactions and their effects upon the current situation, always taking care not to pinpoint the responsibility or guilt of any particular player.
9. From this analysis, players identified some likely paths towards possible solutions (rules, infrastructure, etc.), which they next test by playing new cycles of the game.

The first yearly cycle is used to rectify and validate every participant's understanding and evoke some acknowledgement of other stakeholders' needs and constraints. This first

session thus validates in an explicitly *public* fashion the acknowledgement of the crucial stakes of the situation, detailing competition among uses and the incompatibilities discovered in a realistically operational manner. The seasons, environmental factors (say the degree of flooding) and the locations involved are all taken into account. Afterwards, subsequent sessions naturally lead to productive debate that allows participants *to imagine together solutions that do not somehow punish the "bad conduct" of some stakeholders but improve everybody's satisfaction*. One of the great effects of this kind of self-design process is that people are induced to believe that solutions that are good for everybody are perhaps possible by implementing new rules or infrastructure arrangements. Participants also come to realize that some of the negative aspects of a given situation are not necessarily due to a specific user but were perhaps caused by some elements that simply did not function well together. We may here touch again upon our main hypothesis: the first constraint to the improvement of land and natural resource use management lies more often in socio-political dynamics than in technical progress.

It is worth noting that participants quickly understood the logic of the RPG as well as the needs and incentives of each stakeholder. Take for example the "conflict" that broke out at the game's very first turn between a Park official playing a breeder and a local politician playing a farmer—a type of conflict very frequent in this area (D'Aquino *et al.* 1999, D'Aquino et Seck 2002). The Park official playing a breeder decided to stealthily graze in the Park (strictly forbidden) to satisfy the needs of his herd, then tried to negotiate (without success) access to the Park with his true superior in real life (Who was playing his own role!). In the same way, three peasants playing migratory birds quickly grew to know the needs of wildlife, for example, quietness. After offenses they committed in the first cycle, before they understood the needs of the birds, these peasants were heavily penalized. The peasants eventually came to be the less penalized. And lastly, a hot debate brought Park officials into conflict during the ending assessment. On one side were those who had played farmers and had defended agricultural interests seen as attacked by breeding interests, on the other side were those who had played breeders and had defended the opposite point of view. All this occurred in front of real breeders and farmers, who were quite interested in this turn of events since they themselves had never succeeded in making Park officials agree that these foraging and cropping difficulties existed in the outlying zones of the Park.

Throughout these debates a healthy testing of the rules and infrastructures was implicit that led every participant in the subsequent game sessions to clearly point out the two main causes of difficulties in this case: the lack of *good* rangelands (flooded for two months maximally) because of the Park's and the farmers' unwillingness to resolve the issue of grazing after crops were harvested. Every participant *himself* acknowledged this analysis, even the Park officials and farmers. Hence, debates no longer fruitlessly attempted to fix responsibility and guilt but instead tried to uncover the means ("how?") for handling these matters. This culminated in novel and operationally successful agreements and proposals (see Illustration 5). This type of result was very new in this place where heated tensions existed between breeders and farmers, and between peasants and the Park's administration as well.

At this point the collective progress led people into deeper dialogues that culminated in more accurate planning. Participants finally asked for more convenient forms of *accompanying*, especially by including in the mix the dynamic effects of interactions between the different uses and the medium-term evolutions. We provided them with the necessary supports for a simulation process involving RPG and MAS modeling. This is the second step of our approach.

III. Next, the self-design of a MAS modeling.

Among the range of modeling tools available, Multi-Agent Systems (MAS) seem to be particularly suitable. For several years Multi-Agent Systems have been used in the field of natural and renewable resource management. Our team has designed a special MAS platform, CORMAS (COMmon-pool Resources and Multi-Agent Systems) that greatly simplifies the task of modeling and simulation (Bousquet *et al.* 1998, Bousquet *et al.* 1999), and that is used for knowledge integration in the collective learning processes that focus on integrated natural resource management issues.

Three three-day workshops were organized for the first MAS self-design tests. Around twenty-five people attended each of the workshops. RPG and MAS are indeed particularly suited to complex situations and to prospective studies, being themselves complex systems involving random and somewhat unpredictable dynamics. Actually, RPGs are themselves models of real situations, and it is much closer to reality and easier for people than computer modeling. However, RPGs are often rather cumbersome at the operational level and for providing continuous support in an ongoing decision-making process. Moreover, RPGs do not allow for a sufficiently incremental and iterative process that progressively integrates information and knowledge. Therefore, we used RPGs as a kind of an interface between computer modeling (GIS and MAS) and people (users, stakeholders, policy makers and field experts), allowing them to participate in the design of the model as well as for facilitating their own use of the simulations (Barreteau et Bousquet 2000, Bousquet *et al.* 2001, Barreteau *et al.* 2001, Boissau and Castella 2001, Bousquet *et al.* 2002)⁴. RPGs are thus able to simulate scenarios that are imagined by participants and to trigger group discussion of possible interactions between users and resources. While GIS allows a more complex integration of information and MAS explores more rapidly and systematically the richer and incremental processes and controls involved, RPGs are useful for supporting dialogues between people and computer models. Based on this reasoning, in the following SelfCormas experiments we have tested RPG as a progressive way of designing novel computer models (GIS and MAS) more suitable for resolving the issues and perceptions of the people involved.

Exhausted by long game sessions, people eventually come to ask for a more convenient form of support. Computer modeling (for us, the RPG is a model itself) then becomes worthwhile. Thus we *accompany* players towards a CORMAS platform that adopts the previous role-playing game model: a *MAS self-design*. The same setting of the game, the

⁴ We have experimented with the joint use of RPG and MAS in several kinds of situations (collective irrigation systems, decentralized territory management, multiple uses of forests, etc.). See <http://cormas.cirad.fr>.

same GIS maps, and the same crude rules designed by participants are transferred from the game into a MAS model. Our objective is to test a self-designed MAS requiring modelers to concentrate on producing an upstream modeling environment, rather than a real model, involving as little prior design work by the modeler as possible and enabling stakeholders to express themselves right from the initial stages in the design of their own model. In fact, in our previous self-designed RPG, modelers, who attended the RPG as mere spectators of the discussions, had transcribed the crude rules of the RPG into a MAS model. They had also amended the platform so that the model could test the kinds of changes and regulatory scenarios imagined by participants. Thus, the multi-agent model corresponding to the RPG was programmed from the self-designed RPG (see the last picture in Figure 4). In other words, we let people construct and then test a model through a RPG before providing them with a computer-modeling platform.

Since this flexibility in CORMAS allows us to model exactly the previous RPG features, participants have no problem in understanding the MAS model and they immediately launch into simulation (see Illustration 6). By progressing gradually over a short time, the stakeholders learn to follow the different scenarios on screen, discuss them and propose new ones. After playing out the scenario identified in the previous RPG, new situations emerge, are simulated and then discussed. This provides an opportunity for testing the sensitivity of a given set of collective rules and their consequences with respect to the shared set of assumptions already involved in the behavior of the individual stakeholders. Hot debates may often emerge and demonstrate the worth of these tools for the improvement of the collective processes that we hope to foster (see Figure 5). We strive to keep open additional possibilities for going further than we do with the game sessions alone. This approach has been used at land-use scales from a few hundred square kilometers to 2,500 km² during the same time period. In fact, all this flexibility leads to an incremental, progressive and iterative set of supports to accompany the complex issues of land-use and self-management. ***In our experiments the model has grown progressively richer at the same rhythm as has the collective decision-making process,*** thanks to its use in different situations and at different scales as well as its ongoing progress at a given site. Our incremental and bottom-up accompanying has always created increasing progress towards evermore complex forms of self-government.

IV. Conclusions and perspectives.

Developing RPG in conjunction with GIS and MAS is an effective and valuable way of transferring a large measure of the hands-on power to the users. It would have been physically impossible without computer simulation to play out the different scenarios selected by the stakeholders and to observe their multiple impacts over sufficiently long time intervals. A modeling platform that is sufficiently flexible offers many more possibilities for modifying the rules on demand than do cumbersome game or workshop sessions. The participants were themselves the initial designers of the models, and they were entirely aware of the gap between themselves and reality. Focusing on dynamics instead of results (MAS), and focusing on wide-ranging analysis instead of quantitative data (GIS), are the primary ways in which modeling results should be used. Having largely designed the model, participants were well aware that it was not a “magic” black

box capable of seeing into the future. Thanks to the self-design aspect of the project participants became capable of having more and more insights into the results of the simulations. The stakeholders who developed and played the game were fully capable of interpreting the results of the model and linking them with reality. Even the high rate of illiteracy and lack of formal education among the participants was not an obstacle to the use of computer simulations with this method. Moreover, the self-design approach enabled participants to not lose the key advantages of the RPG during the transfer to the computer models. In effect, RPGs have a specific advantage that computer models that are not self-designed lack: they may benefit from the players' own personal history (Ginty 2000), which is an interesting feature of dynamic modeling. However, our self-design approach should effectively alleviate this difficulty encountered with non self-designed models.

Moreover, in order for the main land and resource management problems to easily emerge in the simulations, there has been no need for the use of sophisticated agents. As the stakeholders themselves instigated the simplifications, they reflect the most important aspects of their actual behavior. Simulation thus multiplies the effectiveness of collective and participatory thinking and is able to take the decision-making process much farther. Currently, we envision the use of computer modeling as a group-decision support tool that will improve negotiation.

Last, but not the least as regards our aims, our experiments enabled new progress to be made in negotiations in a thoroughly conflict-laden local context. After our experiments, many practical achievements in regional planning have been inspired from the use of these supports (D'Aquino *et al.* 1999, D'Aquino et Seck 2002). For instance, in one of the experiments, the establishment of protected rangelands was debated, which had previously been a taboo subject, and a decision was even made to ask the local technical services to talk about the possibilities of improving rangeland yield in these new protected areas. In another area, in which the problem was access to water, simulation resulted in an agreement on new collective regulations that allowed water access for all activities. This process also begins to change the behavior of local stakeholders and principals with respect to their technical partners. Thanks to this supportive modeling becoming both an endogenous activity as well as technically valuable, the representatives of local populations, who themselves identified the technical advice they required within their decision-making process (information on rangeland development, appraisals of certain crops on sandy soils, etc.), then directly contacted the operators of technical services to satisfy their needs. Thus, self-design methods allowed participants (whether stakeholders, principals, or local politicians) to progressively handle all these tools in a more empowered manner in the sphere of local land-use management.

In conclusion, regarding our theoretical assumptions, we seek to establish a supportive framework for dialogue about possible outcomes rather than to impose a specific resolution for complex problems. In line with our opinions, it is not up to the model to provide solutions to problems, but to encourage the discussion of the different alternatives available and to improve the effectiveness of a collective decision-making process (including changes within institutions). It entails putting modeling at any actor's

disposal whatever the model's mental representation may be and promoting simulations which may be initialized with several formalized rules, and then regularly enriching the process in light of the outcome of the previous simulations. This self-incremental design and use of the model allows participants to tackle a complex situation through successive iterations, grasping its most essential dynamics rather than identifying through the modeling so-called "good" solutions. This paves the way for an incremental and iterative understanding of the complexity involved and for the creation of self-supplying institutions (Ostrom 1990) that advance by trial and error methods practically and step-by-step towards better self-government. Progress in decision-making and institutional improvement from within are in line with our efforts in these respects also.